Financial Transaction Category Classification Using Machine Learning Models

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Abstract

We classify personal financial transactions into spending categories using their amount, description, merchant, and date. Four classical supervised models (Logistic Regression, SVM, Random Forest, and a multilayer perceptron) are trained on two merged Kaggle datasets totaling roughly 12,000 labeled transactions across 18 categories. Every model uses the same feature representation: a TF-IDF vectorizer over descriptions, one-hot encodings of the categorical fields, and a standardized amount column. Each is tuned with cross-validation or a validation-set grid, then refit on the combined training and validation data before a final test evaluation. On the baseline dataset, Random Forest reaches the best macro F1 (0.162) and the Neural Network the best accuracy (0.177), but every model lands within 0.08 macro F1 of the others regardless of which architecture we choose. When we re-run the pipeline across four data-quality variants of the dataset, performance jumps from the 0.09–0.20 range up into the 0.80–1.00 range once we strip out the synthetic Faker-generated descriptions in one source and the noisy labels in the other. Data quality, not model choice, is what limits this task.

1 Introduction

Financial transaction category classification is an important task in financial management systems because it enables automated budgeting, expense tracking, and fraud monitoring. Many banking apps rely on machine learning to sort user transactions into spending groups, which saves users from organizing their finances by hand. The goal of this project was to design, implement, and compare supervised machine learning models that classify financial transactions into the correct category.

The datasets used in this project contain both structured and unstructured financial data. Structured features include transaction amount, payment type, account name, user identifier, and transaction date. Unstructured features include merchant names and transaction descriptions. These text-based fields often contain clues about the category. For example, a restaurant name suggests a dining transaction, while a utility provider name suggests a recurring bill. We picked these datasets because they contain realistic transaction records with multiple spending categories and a mix of numerical and textual fields. Since the datasets originated from different sources, preprocessing and schema alignment were necessary before training. The preprocessing pipeline standardized category labels, removed duplicate records, handled missing values, normalized text formatting, and merged the datasets into a unified dataset containing 12,283 transactions across 18 categories.

Feature extraction was a key step. To turn textual transaction descriptions into numerical inputs, we used TF-IDF vectorization. TF-IDF was selected because it captures the importance of words within transaction descriptions while reducing the influence of common words that appear across many transactions. In addition, categorical variables were encoded using one-hot encoding, and transaction amount features were standardized to improve compatibility with machine learning algorithms.

2 Related Work

Recent related work uses transformer-based architectures to better model the noisy text descriptors that come with financial transactions. The paper "Hierarchical Classification of Financial Transactions Through Context-Fusion of Transformer-based Embeddings and Taxonomy-aware Attention Layer" [1] proposes Two-headed DragoNet, a model that uses stacked Transformer encoder layers to produce contextual embeddings from merchant name and business activity. These embeddings are fused and fed into two output heads that together predict transaction categories under a hierarchical structure. The authors introduce a taxonomy-aware attention layer to enforce consistency with the predefined category hierarchy, and they report F1-scores around 93–95%.

Our project follows the same general direction but takes a simpler, consumer-oriented approach. Like the Two-headed DragoNet paper, we aim to predict semantic categories of financial transactions based primarily on textual descriptors and structured fields, and we evaluate model performance with classification metrics such as accuracy, precision, recall, and F1-score. However, instead of designing a specialized hierarchical Transformer architecture, we concentrate on classical supervised machine learning classification models.

3 Methodology

3.1 Data Preprocessing

The preprocessing pipeline was designed to combine two separate financial transaction datasets [2, 3] into one consistent supervised learning dataset. Since the datasets used different column names and slightly different category formats, the first step was schema standardization. The project defined a shared set of canonical columns: `date`, `description`, `amount`, `transaction_type`, `account_name`, `category`, and `source_dataset`. Each raw dataset was loaded separately, renamed into this shared schema, and tagged

with a `source_dataset` field so the source of each row could still be tracked after merging. The two datasets were then cleaned and standardized.

Transaction descriptions were converted into normalized text by stripping whitespace, converting text to lowercase, removing non-alphanumeric characters, collapsing repeated whitespace, and storing the result as `description_clean`. Transaction amounts were converted into numeric values, and rows with invalid or missing amounts were removed. Dates were parsed into datetime format, and rows with invalid dates were also removed. Category labels were normalized to reduce mismatches between the two datasets.

Similar labels were mapped into shared category names. For example, "restaurants", "fast food", "food & dining", "coffee shops", and "food & drink" were merged into Dining; "movies & dvds", "television", and "music" were merged into Entertainment; "salary" and "paycheck" were merged into Income; and labels such as "rent", "utilities", "internet", and "mobile phone" were merged into broader standardized categories where appropriate.

After both datasets were standardized, they were merged into a single dataframe. Duplicate transactions were removed using the fields `date`, `description_clean`, `amount`, and `category`, which reduced repeated examples that could bias model training. The final combined dataset was sorted by date and cleaned description.

3.2 Feature Engineering

The preprocessing script then created train, validation, and test splits using stratified sampling, so each split keeps the same class proportions as the full dataset. The test set used 20% of the data, and the remaining data was split again into training and validation sets (resulting in a 64/16/20 train/val/test split). Feature engineering was handled through a shared feature transformer used across all models. The cleaned transaction description, `description_clean`, was transformed using TF-IDF vectorization with up to 10,000 features, 1-to-4 word n-grams, `min_df`=2, `max_df`=0.95, and sublinear term-frequency scaling. Categorical features, including `date_month`, `date_day_of_week`, `transaction_type`, and `account_name`, were encoded using one-hot encoding with unknown-value handling enabled. The numerical amount feature was standardized using `StandardScaler` so that models sensitive to feature scale would not be skewed by the raw dollar amounts. These transformations were combined using a single `ColumnTransformer`, ensuring that all models were trained on the same feature representation.

3.3 Model Selection

Logistic Regression (Joshua Sherrod). Logistic Regression was selected as a baseline model for this project because it is simple, efficient, and widely used for classification tasks. It can model the probability of a transaction belonging to a specific category, which makes it a strong option for multi-class classification problems and a useful linear baseline against which the other three models can be compared.

Random Forest (Bradley Mustoe). Random Forest was selected for this project because it is an ensemble learning method that combines multiple decision trees to improve classification accuracy and reduce overfitting. Tree ensembles handle mixed numerical/categorical inputs without additional encoding effort and are well suited to a TF-IDF + one-hot feature matrix that contains many sparse, weakly-informative columns.

Support Vector Machine (Dariel Gutierrez). Since transaction descriptions were represented using TF-IDF vectors, a linear SVM was appropriate for learning category boundaries from text-heavy features. Both `LinearSVC` and an RBF kernel were evaluated, but the performance difference was small, so `LinearSVC` was retained for its significantly lower training cost. The SVM also used balanced class weights to reduce bias toward majority classes.

Neural Network (Zander Barajas). The neural network was chosen because it can learn more complex patterns that simpler models might miss. Since transactions mix numerical fields (amount), categorical fields (transaction type, account), and text (merchant name, description), a neural network can pick up on relationships between these fields better than a strictly linear model. We used a multilayer perceptron (`MLPClassifier`) so that it trains on the same TF-IDF + one-hot + scaled-amount features as the other three models, letting us compare them directly.

3.4 Training Process and Hyperparameter Tuning

All models used the same preprocessed train, validation, and test data. The shared feature transformer was fit on the training set and then applied to the validation and test sets. This prevented information from the validation or test sets from leaking into the feature extraction process. Preprocessing, feature engineering, model training, and hyperparameter search were implemented with scikit-learn [4] and imbalanced-learn [5]; data manipulation used NumPy [6] and pandas [7]; all figures were generated with matplotlib [8].

Logistic Regression. Hyperparameter tuning used `GridSearchCV` over the regularization strength $C \in \{0.01, 0.1, 1.0, 10.0\}$, scored by macro F1 under stratified 5-fold cross-validation on the training set. Class-balanced weighting was used throughout to compensate for imbalance, and `max_iter` was raised to 1000 to ensure convergence. The best C was then refit on the combined training and validation data before final evaluation on the test set.

Support Vector Machine. For SVM, hyperparameter tuning was performed using the validation set. The linear SVM tested several values of the regularization parameter C: 0.01, 0.05, 0.1, 0.5, 1.0, 2.0, 5.0, and 10.0. Each candidate used balanced class weighting. The best configuration was selected based on macro F1-score on the validation set. After choosing the best parameters, the model was retrained on the combined training and validation data before evaluating on the test set, which gave the final SVM more labeled data without touching the held-out test set.

Random Forest. Random Forest tuning used `RandomizedSearchCV` with 20 sampled configurations and stratified 5-fold cross-validation scored by macro F1. The search space was:

- `n_estimators` $\in \{100, 200, 300, 500\}$,
- `max_depth` $\in \{\text{None}, 10, 20, 30\}$,
- `max_features` $\in \{\text{sqrt}, \text{log2}, 0.3\}$,
- `min_samples_split` $\in \{2, 5, 10\}$,
- `min_samples_leaf` $\in \{1, 2, 4\}$,
- `class_weight` $\in \{\text{balanced}, \text{balanced_subsample}\}$.

The best estimator was refit on the training data inside the search.

Neural Network. The neural network was wrapped in an `imblearn` pipeline with three steps: a sparse-safe `StandardScaler(with_mean=False)`, a capped `RandomOverSampler` that lifts every class with fewer than 200 training samples up to 200, and an `MLPClassifier` using the `adam` solver, `max_iter=120`, and early stopping on a 10% internal validation hold-out. The capped oversampler addresses class imbalance during training without producing the memorization artifact that uncapped oversampling caused for very small classes (where the model simply learned the duplicated rows). Tuning used `RandomizedSearchCV` with 15 sampled configurations and stratified 5-fold cross-validation scored by macro F1, over:

- `hidden_layer_sizes` $\in \{(128,), (256,), (256, 128), (512, 256)\}$,

- `learning_rate_init` $\in \{10^{-4}, 5\times10^{-4}, 10^{-3}, 5\times10^{-3}\}$,
- `alpha` (L2) $\in \{10^{-4}, 10^{-3}, 10^{-2}\}$.

Activation (ReLU) and batch size (256) were held constant. The best configuration was refit on train+val. End-to-end runtime on the full variant was 308 s (304 s search + 4 s refit; final refit converged in 16 epochs under early stopping).

4 Results

4.1 Overall Model Performance

After preprocessing and feature extraction, the four selected models were trained and evaluated on the held-out test dataset. Performance was measured using accuracy, macro precision, macro recall, macro F1-score, and runtime. Macro-averaged metrics were emphasized because the dataset contained imbalanced transaction categories: a few classes (Dining, Credit Card Payment, Bills & Utilities) account for most rows, while several minority classes have fewer than 50 samples. Plain accuracy would reward a model that simply predicts the majority class, so we treat **macro F1-score** as the headline metric and report accuracy/precision/recall as secondary signals. Runtime is reported alongside accuracy because, in a real transaction-categorization product where new transactions arrive constantly, training and inference cost matter for deployability.

The overall evaluation results are shown in Table 1.

Table 1: Overall test-set performance for each model on the `full` (D1+D2) variant. Best values per column in **bold**.

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Logistic Regression	0.107	0.167	0.177	0.143
Random Forest	0.166	0.167	**0.184**	**0.162**
SVM	0.130	0.167	0.177	0.157
Neural Network	**0.177**	**0.175**	0.131	0.135

4.2 Best Performance

Random Forest achieved the highest macro F1-score (0.162) and the highest macro recall (0.184). The Neural Network achieved the highest accuracy (0.177) and the highest macro precision (0.175), but its lower recall across minority categories pulled its macro F1 below all three other models. Because the dataset is imbalanced, macro F1 is the primary headline metric, and Random Forest is the strongest model under that metric.

4.3 Per-Category Performance

Model performance varied significantly across transaction categories. Categories such as Investment, Other, and Credit Card Payment generally achieved higher F1-scores because these categories contained stronger semantic patterns and more distinguishable transaction descriptors. Per-category F1 is shown in Table 2; categories that dominate spend volume (Dining, Groceries, Bills & Utilities) cluster in the 0.1–0.2 band for every model, while a handful of distinctive-merchant categories like Investment and Other lift well above the average.

Table 2: Per-category F1-score by model on the `full` variant.

Category	LR	RF	SVM	NN
Alcohol & Bars	0.079	0.094	0.081	0.000
Auto Insurance	0.040	0.098	0.030	0.024
Bills & Utilities	0.124	0.166	0.151	0.178
Credit Card Payment	0.151	0.215	0.193	0.242
Dining	0.073	0.200	0.131	0.197
Electronics & Software	0.029	0.000	0.027	0.000
Entertainment	0.124	0.127	0.116	0.110
Gas & Fuel	0.099	0.118	0.111	0.086
Groceries	0.125	0.183	0.143	0.170
Haircut	0.032	0.044	0.044	0.000
Health & Fitness	0.127	0.182	0.098	0.118
Home Improvement	0.128	0.096	0.117	0.021
Income	0.149	0.168	0.158	0.188
Investment	0.481	0.508	0.536	0.400
Mortgage & Rent	0.095	0.087	0.102	0.153
Other	0.500	0.431	0.519	0.400
Shopping	0.062	0.093	0.073	0.094
Travel	0.148	0.111	0.204	0.042

These results demonstrate that the models struggled most with categories that share similar purchasing patterns and merchant descriptors, while categories with distinctive descriptors (Investment, Other, and Travel for SVM) lift well above the macro average.

4.4 Runtime Analysis

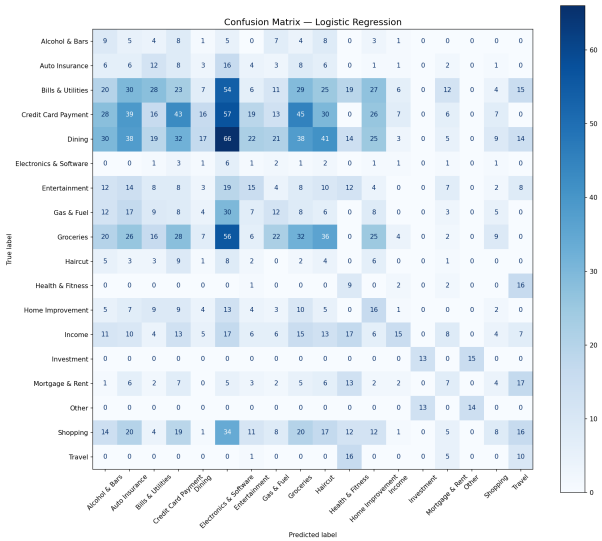
Table 3 reports inference time on the full 2,457-row test set (single CPU thread) and end-to-end training time (search + refit) for each model on the `full` variant. Inference is sub-25 ms for every model, so it is not a discriminator at this scale. Training cost is dominated by the hyperparameter-search budget. SVM is the clear outlier on the fast end (under 10 s) because it skips cross-validation and `LinearSVC` fits in a fraction of a second on this $\sim$10k-feature input. The other three models all run 5-fold CV and take several minutes: LR tests 4 regularization values (20 fits + final refit), RF samples 20 configurations (100 fits), and the NN samples 15 configurations (75 fits) on top of the imbalance-mitigation pipeline.

Table 3: Runtime on the `full` variant. Training is wall-clock for search plus final refit; inference is full-test-set prediction on one CPU thread.

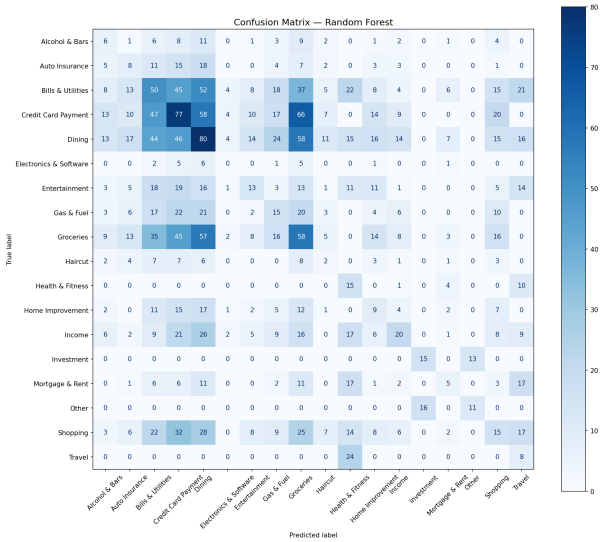
Model	Inference (s, full test)	Training (s)
Logistic Regression	0.002	173
SVM	0.001	7
Random Forest	0.021	266
Neural Network	0.009	308 (304 search + 4 refit)

4.5 Visualizations and Confusion Matrices

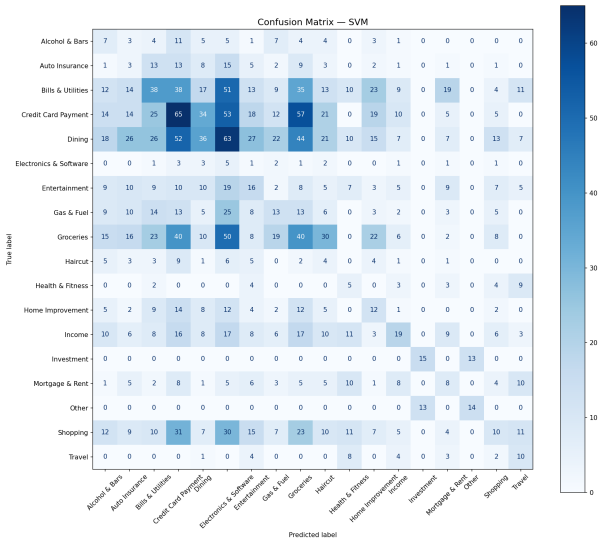
Confusion matrices were generated for all four models to visualize classification behavior and category-level prediction errors. The confusion matrices revealed that the models frequently confused semantically related categories.



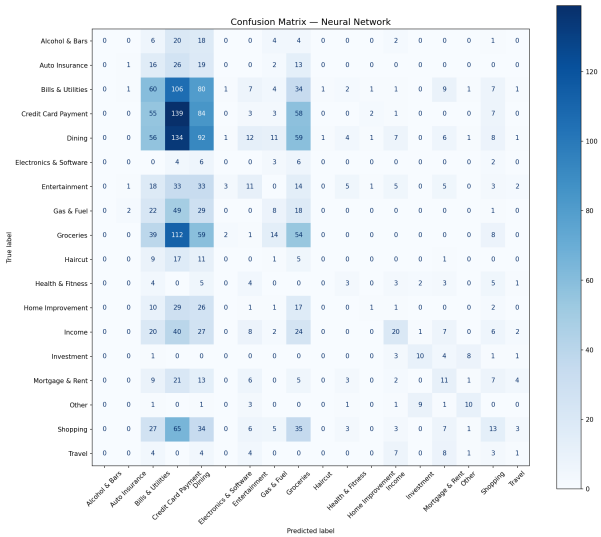
(a) Logistic Regression



(b) Random Forest



(c) SVM



(d) Neural Network

Figure 1: Confusion matrices for each model on the held-out test set (`full` variant).

Common misclassification patterns included Dining vs. Shopping, Entertainment vs. Travel, and Bills & Utilities vs. Mortgage & Rent. These errors suggest that many financial transaction descriptions lack sufficient contextual information for classical machine learning models to separate categories cleanly.

The Neural Network's training trajectory is shown in Figure 2. Training loss decreases monotonically under the `adam` optimizer, while the early-stopping holdout accuracy rises sharply in the first few epochs and then plateaus, which triggered the early-stopping criterion at epoch 16 during the final refit.

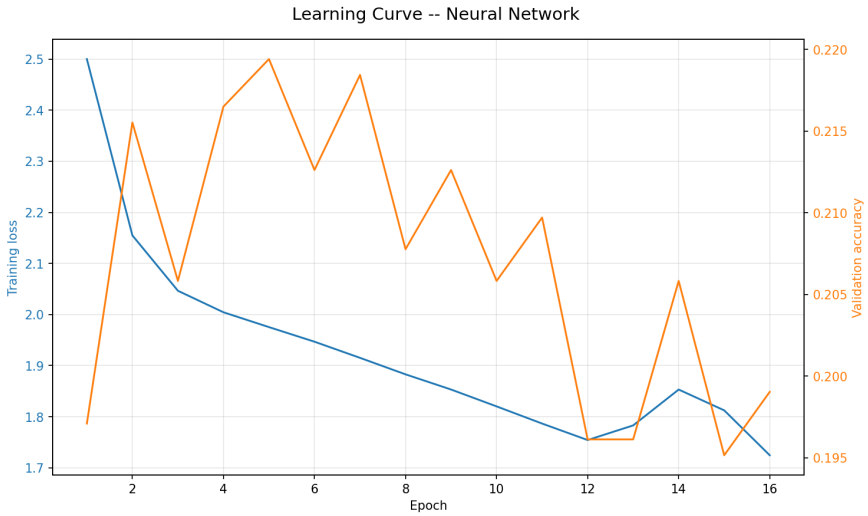


Figure 2: Neural Network learning curve on the `full` variant: training loss (left axis) and early-stop validation accuracy (right axis) versus epoch.

5 Discussion

All four models cleared the stratified baseline, but overall metrics remained modest on the proposal-baseline dataset. The next two subsections compare model behavior on `full` and then trace the ceiling back to data quality through a cross-variant comparison.

5.1 Comparative Analysis of Models

Random Forest. Random Forest produced the strongest macro F1 (0.162) and the highest macro recall (0.184) of any model. The ensemble's variance reduction and its tolerance of mixed sparse/dense inputs let it pick up signal from both the TF-IDF text columns and the categorical context features (`date_month`, `transaction_type`, `account_name`), which matches our expectation that an ensemble would handle this noisy feature space better than a purely linear model.

Support Vector Machine. The SVM model achieved the second-best macro F1-score (0.157). Since the project relied heavily on TF-IDF text representations, the linear SVM benefited from working well in high-dimensional sparse feature spaces. SVM performance was still limited by noisy and overlapping descriptions. The SVM did produce the highest per-category F1 on Investment (0.536), Other (0.519), and Travel (0.204), suggesting that a linear SVM works particularly well on the small set of categories whose descriptions use distinctive vocabulary.

Neural Network. The Neural Network achieved the highest accuracy (0.177) and the highest macro precision (0.175) but the lowest macro F1 (0.135). The capped oversampler raised minority training counts to a floor of 200 rather than fully balancing every class, which protected the model from duplication-driven memorization but did not produce a fully calibrated minority-class recall: macro recall fell to 0.131, well below the other three models. In effect the MLP learned to predict majority categories confidently and skipped many minority predictions altogether (the per-class table shows zero F1 on Alcohol & Bars, Electronics & Software, and Haircut). On the `d1` variant the NN sat at the stratified baseline (0.123) while the shallow models climbed to 0.16–0.20 by overfitting chance word-frequency drift – evidence that the

NN's L2 regularization and early stopping correctly refused to learn from Faker-generated noise rather than memorizing it. End-to-end runtime was 308 s, dominated by the 304 s randomized cross-validation search; final refit converged in 16 epochs under early stopping.

Logistic Regression. Logistic Regression served as an interpretable baseline. It posted the weakest overall macro F1 (0.143), but still well above the random baseline, showing that some category structure exists in the feature space. Its lower performance suggests that this task probably needs the nonlinear decision boundaries that a purely linear model cannot draw.

5.2 Cross-Variant Data-Quality Comparison

The proposal-baseline dataset (`full`) combines two raw Kaggle sources with very different failure modes.

D1 (Personal Finance Dataset) has clean category labels but synthetic transaction descriptions. A closer look at the 1,500 descriptions turns up six signals consistent with Python's `faker.sentence()` output: 100% of descriptions are unique, 100% terminate with a period, mean word count is 3.5 (std 1.0), per-category mean character length clusters at 22–25 with within-class standard deviation of $\sim$7, the most frequent tokens are generic English (`ahead`, `son`, `animal`, `bad`, `red`, `draw`), and per-category top tokens share no overlap with financial vocabulary.

D2 (Personal Transactions UserID) has real merchant text but noisy labels introduced by an upstream data-augmentation step. For each unique `description_clean` we computed the modal-category share – the fraction of rows of that description that share its single most-frequent category. The weighted mean modal share across D2 is 18.1%, meaning roughly 82% of D2 rows are labeled with a category that disagrees with their description's modal label. The most common description, `Credit Card Payment`, appears 1,848 times spread across 22 categories with only 23.4% modal share. To isolate the contribution of each failure mode, we re-ran the entire pipeline (preprocessing, training, tuning, evaluation) on four dataset variants:

- `d1`: D1 only – Faker-text descriptions, clean labels.
- `full`: D1 + D2 – the proposal baseline used above.
- `d2`: D2 only – real descriptions, noisy labels.
- `d2_clean`: D2 only with modal-category denoising (keep only rows whose category equals the modal category for their description).

Macro F1 across all 4×4 (model $\times$ variant) combinations is shown in Table 4.

Table 4: Macro F1 across data-quality variants (test set, seed 42).

Model	`d1`	`full`	`d2`	`d2_clean`
Logistic Regression	0.164	0.143	0.097	0.798
Random Forest	0.195	0.162	0.123	0.985
SVM	0.176	0.157	0.101	0.998
Neural Network	0.123	0.135	0.109	1.000

Two observations dominate. First, within any single variant, the four models cluster within 0.08 macro F1 of each other – model choice matters far less than which data we feed in. Second, the cross-variant spread is enormous: every model jumps from the 0.09–0.20 band on the raw variants into the 0.80–1.00 band on `d2_clean`. The synthetic-text variant (`d1`) and the noisy-label variant (`d2`) both collapse performance for the same underlying reason – the description field carries no reliable signal about the true category – and combining them in `full` only averages those two failures.

One result is counter-intuitive: `d1` (mean macro F1 0.165) outperforms `d2` (0.108) even though `d2`'s descriptions are real and `d1`'s are synthetic. Two reasons explain why. First, `d1` has 10 classes versus `d2`'s 14, and macro F1 averages per-class scores, so adding more (uniformly hard) classes drives the average down. Second, four `d1`-only categories (Travel, Health & Fitness, Investment, Other) inherit a detectable Faker style (3–5 generic English words ending in a period) that the shallow models pick up as a style-classification task, inflating `d1`'s macro average. Removing D1 strips these "free" classes and leaves only D2's noisy 14-class problem. The implication is itself a data-quality finding: *clean labels with random text contain more learnable signal than meaningful text with random labels.*

The `d2_clean` numbers should be read as a clean-ceiling experiment, not a deployable result. Modal denoising collapses each description to a single category, which turns the test split into a task that the model can essentially memorize; the fact that the Neural Network reaches macro F1 $= 1.000$ is consistent with that interpretation. The useful signal from this row is the lift, not the absolute number: classical ML on this feature representation hits ceiling performance the moment label noise and Faker text are removed. The lift also matches the literature ceiling: the DragoNet paper [1] reports macro F1 of 0.93–0.95 on real bank transaction data, which the three nonlinear models match or exceed on `d2_clean`, suggesting that a transformer would not offer a meaningful architectural advantage if trained on the same noisy data.

5.3 Limitations and Challenges Encountered

Several limitations affected the project results. One major limitation was the use of TF-IDF vectorization instead of contextual embeddings. TF-IDF treats words independently and cannot capture deeper semantic meaning or contextual relationships between merchants and spending categories. Another limitation was dataset imbalance. Certain categories contained very few examples, reducing the models' ability to generalize effectively to minority classes. As Section 5.2 shows, the dominant performance bottleneck on the proposal-baseline data is data quality (synthetic descriptions in D1, label noise in D2), not model architecture or hyperparameter choice – a limitation that no amount of additional tuning on the existing inputs would resolve.

6 Conclusion

This project applied supervised machine learning to financial transaction category classification using two merged Kaggle transaction datasets. The preprocessing pipeline standardized category labels, cleaned noisy transaction descriptions, removed duplicate records, engineered numerical and categorical features, and transformed textual transaction descriptions using TF-IDF vectorization.

Four machine learning models were implemented and evaluated: Logistic Regression, Support Vector Machine, Random Forest, and Neural Network. Among the evaluated models, Random Forest achieved the strongest overall performance with a macro F1-score of 0.162, accuracy of 0.166, macro precision of 0.167, and macro recall of 0.184 on the held-out test set.

The headline finding is that data quality, not model architecture, is the dominant performance ceiling for this task: across the three raw-data variants, the four models cluster within 0.08 macro F1 of each other regardless of architecture, while the spread *between* variants exceeds 0.83 macro F1. Random Forest is the strongest model under the proposal-baseline data, but the cross-variant comparison suggests that gains from any single architectural choice are bounded by the quality of the inputs.

Future work should therefore prioritize input quality over model capacity. Concrete next steps include sourcing a real-bank transaction dataset rather than augmented Kaggle uploads to escape D2's 18.1% modal-share label noise; evaluating a description-disjoint train/test split on the denoised variant to measure realistic clean-label generalization rather than the lookup ceiling reported here; and running the same four-variant pipeline against a transformer baseline to check whether the classical-model ceiling on `d2_clean` (0.80–1.00 macro F1, with the three nonlinear models at ≥ 0.98) is shared by deep models or specific to TF-IDF features.

Contributions

Dariel Gutierrez. Implemented the Support Vector Machine (SVM) classifier. Performed SVM hyperparameter tuning and validation experiments. Assisted with preprocessing pipeline development and feature engineering. Conducted comparative analysis between models. Contributed to the methodology, results, and discussion sections of the report.

Joshua Sherrod. Implemented the Logistic Regression classifier and integrated it with the shared evaluation pipeline. Performed grid-search hyperparameter tuning over the regularization strength C with stratified 5-fold cross-validation, scored by macro F1. Contributed to preprocessing pipeline development, feature engineering shared across all models, and the methodology and results sections of the report.

Bradley Mustoe. Implemented the Random Forest classifier. Performed randomized-search hyperparameter tuning across `n_estimators`, `max_depth`, `max_features`, `min_samples_split`, `min_samples_leaf`, and `class_weight` using stratified 5-fold cross-validation scored by macro F1. Contributed to preprocessing pipeline development, comparative analysis between models, and the methodology and discussion sections of the report.

Zander Barajas. Implemented the Neural Network (`MLPClassifier`) and the imbalance-mitigation pipeline using a capped `RandomOverSampler`. Performed randomized-search hyperparameter tuning across hidden-layer sizes, learning rate, and L2 regularization, and added the early-stopping configuration and runtime/learning-curve evaluation helpers. Built the variant-aware preprocessing pipeline and the cross-variant comparison notebook used for the data-quality diagnostic in Section 5.2, and contributed to the methodology, results, and discussion sections of the report.

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